

Organization

Dr. Christian A. Mahringer

TUM School of Management, Campus Heilbronn



Course agenda

Date	Main theme
13.04.2026	Definitional basics of organization
20.04.2026	Formal organizational design
27.04.2026	Organization and environment
04.05.2026	Strategic organization design
11.05.2026	Technology and organization
18.05.2026	Information technology, AI, and organization
01.06.2026	Informal organization

Date	Main theme
08.06.2026	Informal organization
15.06.2026	Dynamic perspectives on organizing
22.06.2026	Guest lecture, Felix Ehrenfels, mmmake
06.07.2026	Trends in organization design
13.07.2026	Trends in organization design
17.07.2026	Review and wrap-up

Session 05

Technology and organization:

From modernist classification to symbolic interpretation

Technologies in organizations

Task:

- What are typical technologies in organizations?
- How do you know it's a technology?

Goal:

- Deepen your understanding of what technology is.

Modus:

- Pair work.

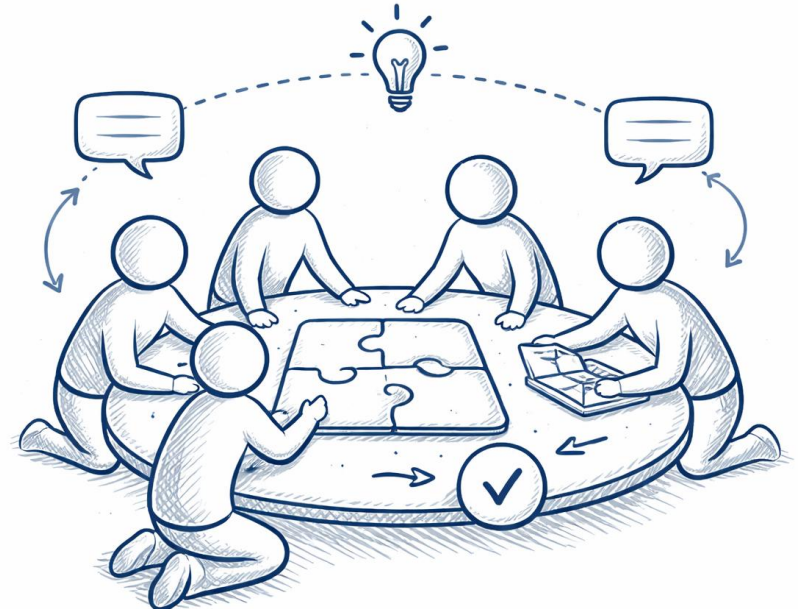


Image source: ChatGPT

What you should be able to do after this session

Learning goals

- Define technology in organizations.
- Analyze technology across task, department, organization, and society.
- Apply Woodward, Thompson, and Perrow to classify technologies.
- Distinguish modernist and symbolic-interpretive views.
- Analyze how technology is enacted differently in practice.



Image source: ChatGPT

Definition of technology

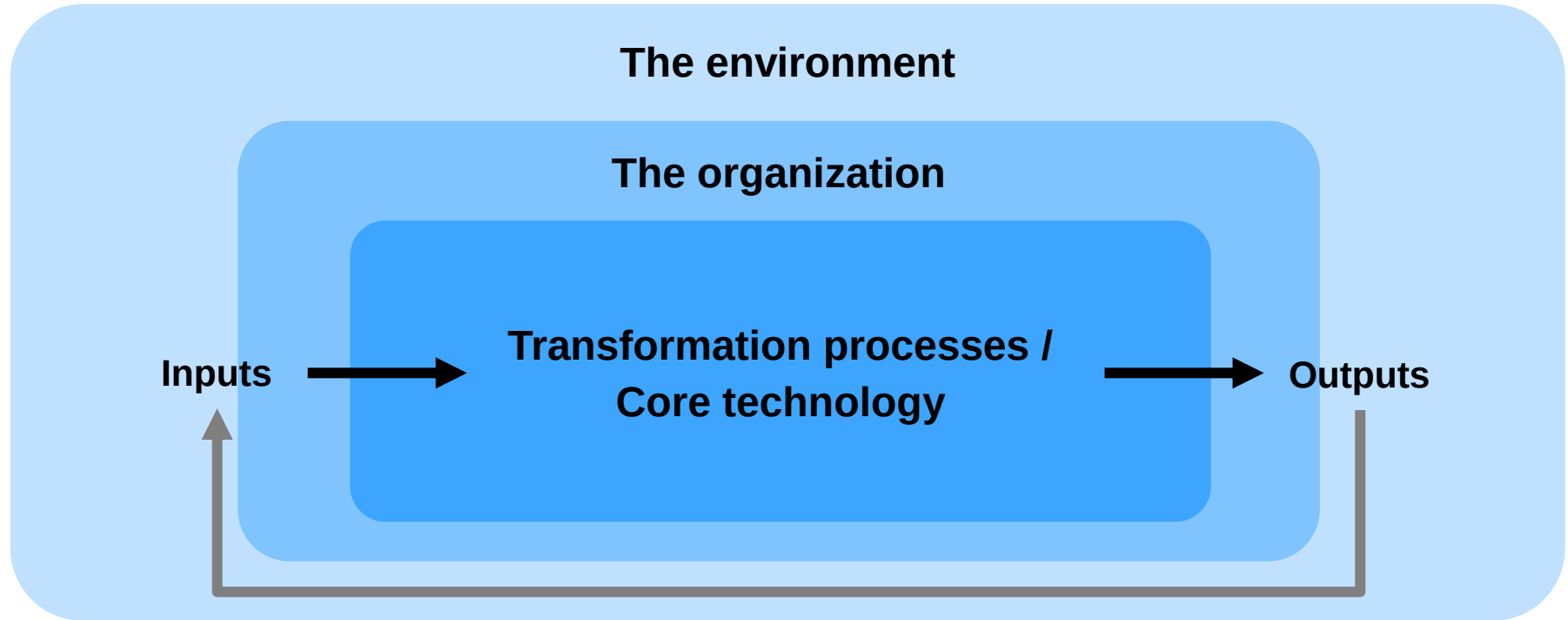
- From Greek technē: art, skill, craft, method.
- From logia/logos: study, knowledge.
- Original meaning: systematic knowledge about arts, crafts, and techniques.
- Modern meaning: (the study and knowledge of) the practical, especially industrial, use of scientific discoveries.
- Technology includes machines, tools, methods, procedures, and skills.
- Historical shift: from craft knowledge to mechanical arts, industrial techniques, and applied sciences.



<https://dictionary.cambridge.org/de/worterbuch/englisch/technology>; <https://medium.com/@mvsudhakaran/technology-the-origin-story-21fda9c08734>;
Image source: pixabay, PixelLabs

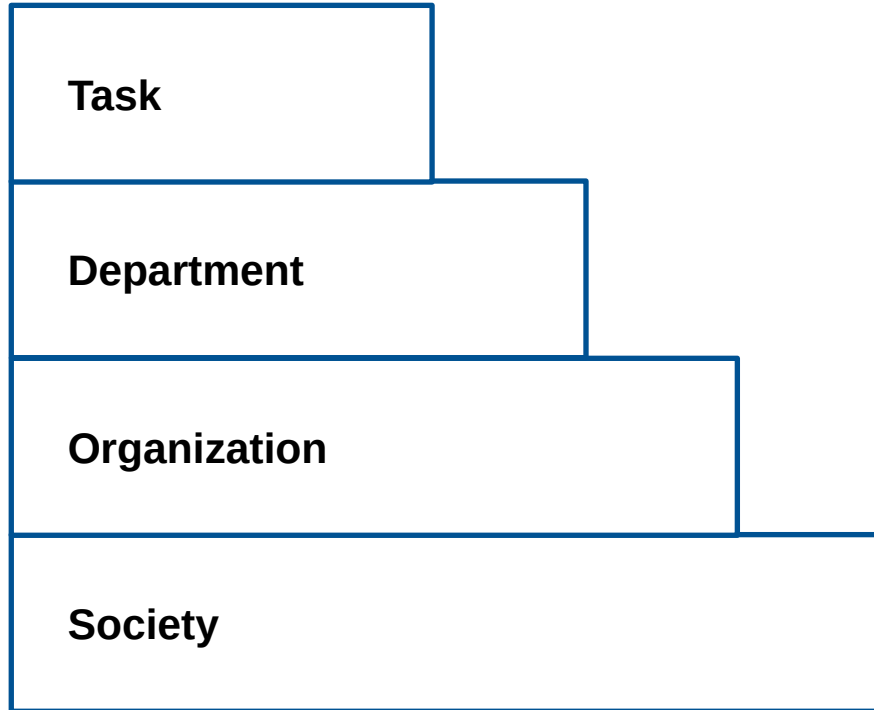
A basic view of organizational technology

Transforming inputs into outputs



Hatch, M. J. (2018). Organization theory: Modern, symbolic, and postmodern perspectives (4th ed.). Oxford University Press.

Technologies can be analyzed on multiple levels



Hatch, M. J. (2018). Organization theory: Modern, symbolic, and postmodern perspectives (4th ed.). Oxford University Press; Image source: pixabay, Alter_Haudegen

Exercise

Task:

- Analyze virtual reality glasses on four levels:
Task, Department, Organization, Society

Goal:

- Understand the broad scope of technology.

Modus:

- Pair work.

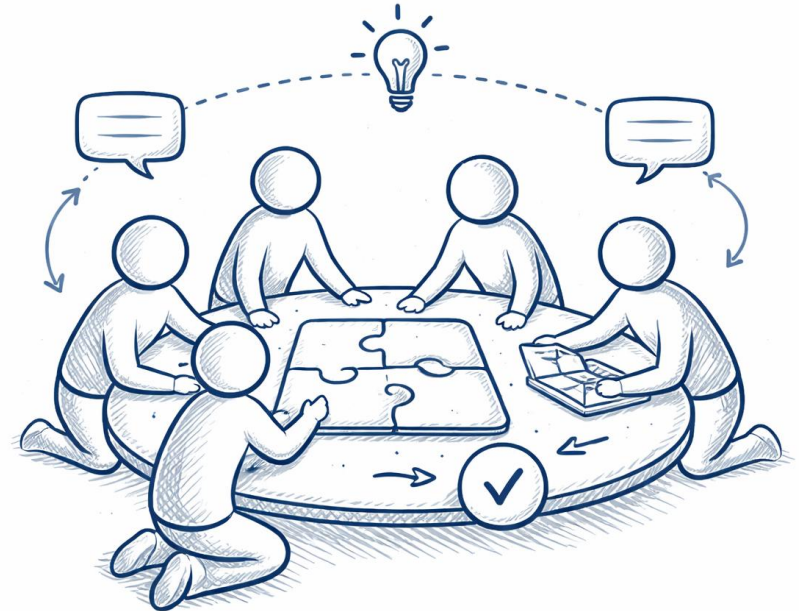


Image source: ChatGPT

Modernist typologies of technology

Three fundamental thinkers

Joan Woodward (1965)

Technical complexity



James Thompson (1967)

Standardization of inputs,
outputs, and processes



Charles Perrow (1972)

Task variability and task
analyzability

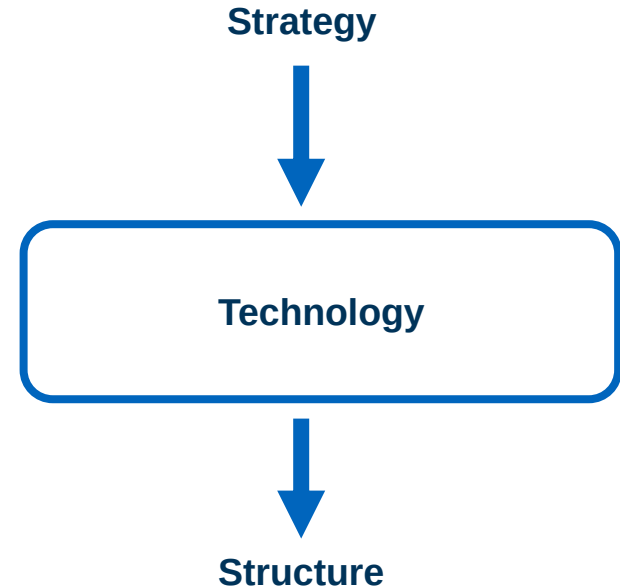


Image sources: amazon.com, alchetron.com, imperial.ac.uk

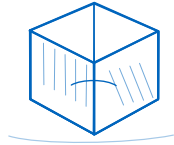
Woodward's starting point

From one best way to contingency

- Woodward originally searched for the one best organizational structure.
- Her finding was different: there is no universally best structure.
- The effectiveness of structure depends on technology.
- Successful organizations align strategy, structure, and technology.



Woodward's technical complexity scale



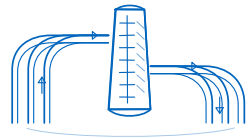
GROUP I
Small batch
and unit
production

- I Production of single pieces to customer orders
- II Production of technically complex units one by one
- III Fabrication of large equipment in stages
- IV Production of pieces in small batches



GROUP II
Large
batch
and mass
production

- V Production of components in large batches subsequently assembled diversely
- VI Production of large batches, assembly line type
- VII Mass production



GROUP III
Continuous
process
production

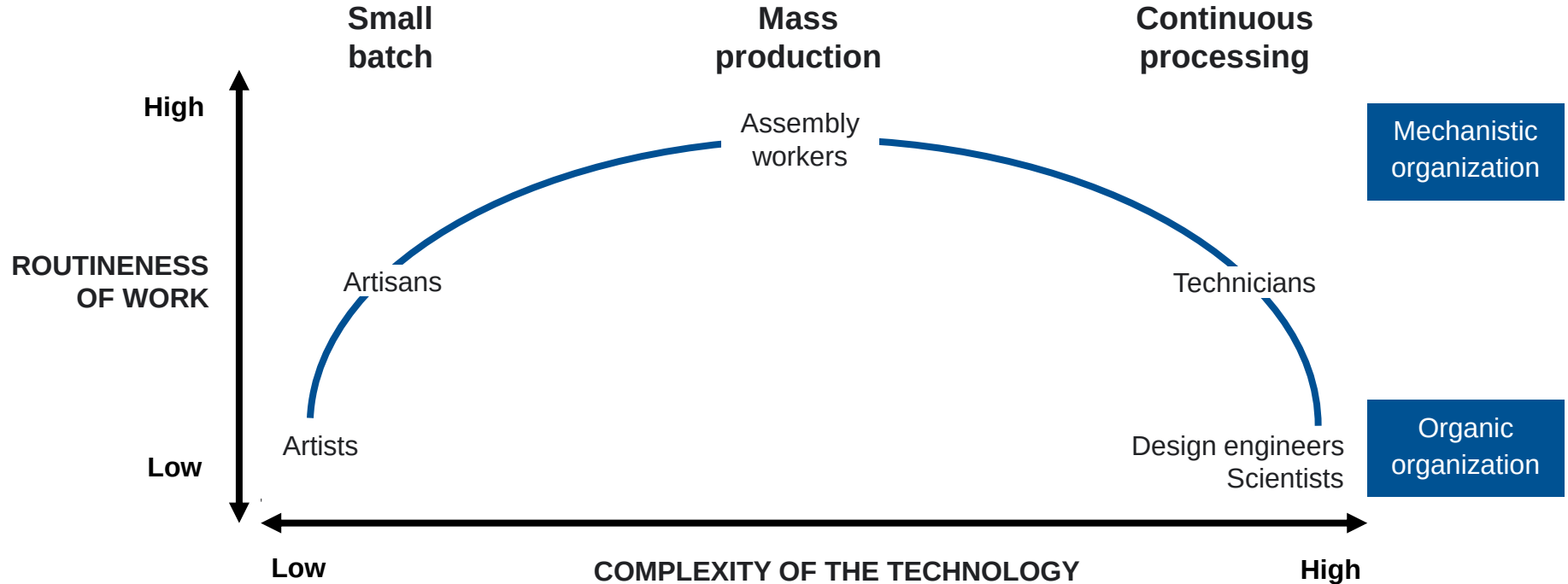
- VII Continuous process production combined with the preparation of a product for sale by large batch or mass production methods
- IX Continuous process production of chemicals in batches
- X Continuous flow production of liquids, gases, and solid shapes

LOW

TECHNICAL
COMPLEXITY

HIGH

Woodward's technical complexity and routinization



Hatch, M. J. (2018). Organization theory: Modern, symbolic, and postmodern perspectives (4th ed.). Oxford University Press

A caution

Manufacturing is not all technology.

- Woodward focused on manufacturing technologies.
- This leaves open how to analyze service organizations.
- Service work often does not produce tangible products.
- Thompson extends the analysis to services.

Thompson's typology

Two forms of organizational technology

Manufacturing technology

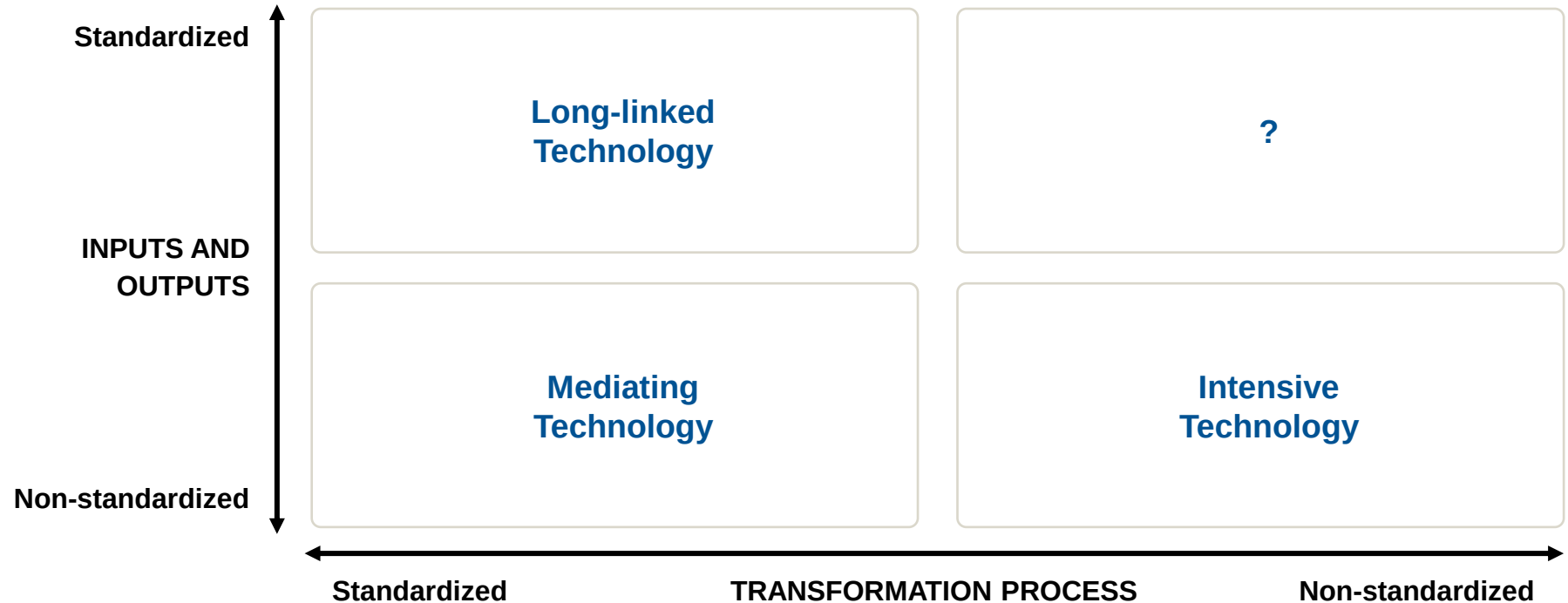
- Tangible product.
- Products can be stored in inventory.

Service technology

- Intangible output.
- Production and consumption occur simultaneously.
- Cannot be stored in inventory.

Thompson's typology

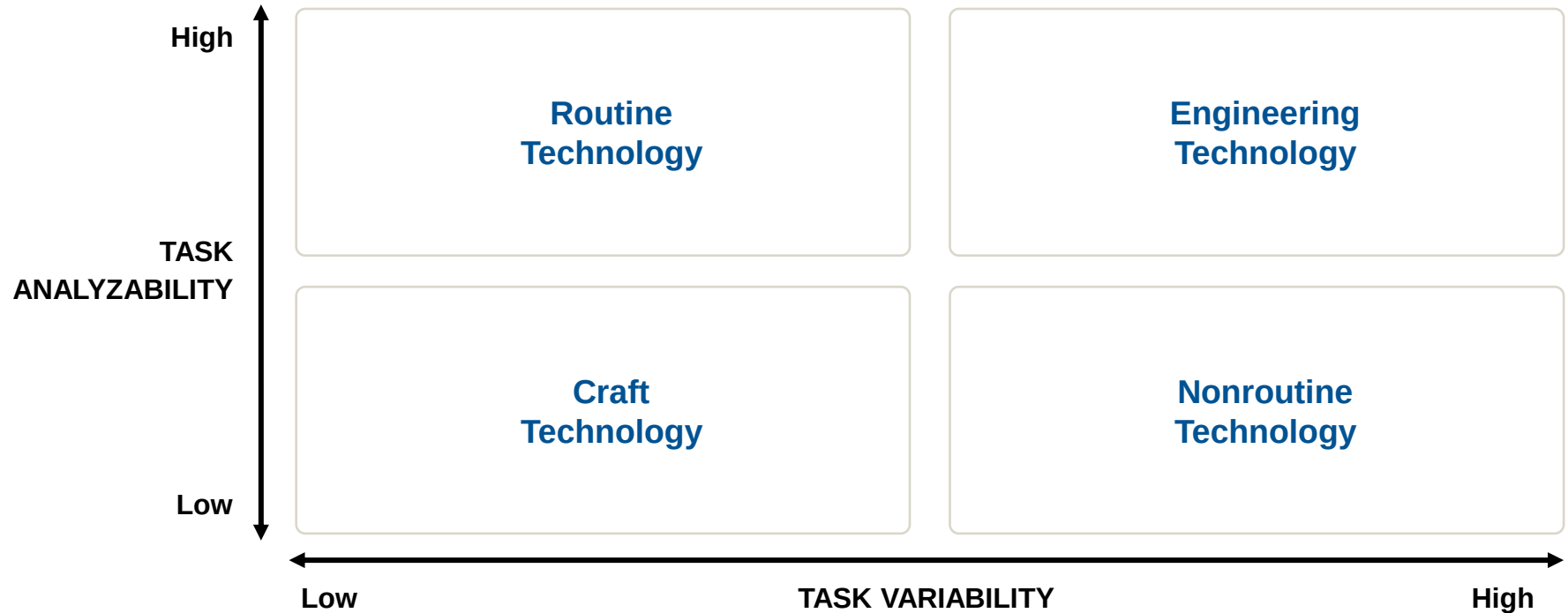
Two forms of standardization classify technologies



Hatch, M. J. (2018). Organization theory: Modern, symbolic, and postmodern perspectives (4th ed.). Oxford University Press.

Perrow's typology

Two forms of standardization classify technologies



Hatch, M. J. (2018). Organization theory: Modern, symbolic, and postmodern perspectives (4th ed.). Oxford University Press.

Synthesis of analytical criteria

Combining Woodward, Thompson, and Perrow

1. Technical complexity (Woodward)
2. Routineness of work (Woodward)
3. Standardization of transformation processes (Thompson)
4. Standardization of inputs/outputs (Thompson)
5. Task variability (Perrow)
6. Analyzability of tasks (Perrow)

Exercise

Categorizing practical applications of technology

Task:

- Analyze one of the following cases according to the analytical criteria on the previous slide.
 - [HOMAG Flat-pack furniture production](#)
 - [Coca-Cola returnable-glass bottling line](#)
 - [Amazon robotic fulfillment center](#)
 - [Nicklaus Children's Hospital](#)
- Craft two slides in [this document](#). One slide to briefly introduce the case and one for the analysis.

Goal:

- Understand technology classifications.

Modus:

- Group work.

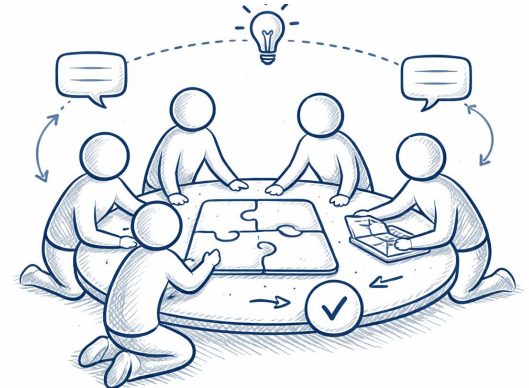


Image source: ChatGPT

Toward the symbolic-interpretative view

From classification to meaning and use

Modernist view: technology as condition

- Technology was treated as a condition that affects structure and coordination.

Symbolic view: technology as enacted in practice.

- Technology is shaped through use.
- Actors appropriate, adapt, and reinterpret technology in practice.
- The same technology can matter differently across contexts.

Social Construction of Technology (SCOT)



- Different social groups make different demands on the same technology.
- In the bicycle case, women cyclists wanted frames compatible with long dresses, while male racers wanted speed and stability.
- Several bicycle designs emerged from these competing demands until market forces selected one stable model.
- Technology and society shape each other, women's fashion influenced bicycle design, and the selected bicycle design later influenced women's clothing practices.
- At the everyday level, users and technicians negotiate the meaning of technology through use, interpretation, and improvisation.
- Orr's Xerox study shows that machines can have both a technical and a social presence in work practices.

Hatch, M. J. (2018). Organization theory: Modern, symbolic, and postmodern perspectives (4th ed.). Oxford University Press.

Adaptive structuration theory

Technology and social structure shape each other

- Technology shapes how people work.
- People shape technology through routines, improvisation, and use.
- The same technology can be appropriated differently by different actors.
- Example: you may use Microsoft Excel very differently than an accountant.

Example of adaptive structuration theory

Incident Tracking Support System (ITSS) in customer support at Zeta

Technology shapes routines

1. Makes work visible

Customer calls and problem-solving steps become documented, searchable, and monitorable.

2. Redistributes responsibility

Support work shifts from individual problem-solving toward shared access and shared responsibility.

3. Enables new coordination patterns

ITSS allows electronic collaboration across specialists, departments, and overseas support offices.

Example of adaptive structuration theory

Incident Tracking Support System (ITSS) in customer support at Zeta

Routines shape technology-in-use

1. Users appropriate features selectively

Specialists use ITSS not only as designed, but adapt it to their own support practices.

2. Users develop new routines around the system

Specialists create routines for documenting, reusing, transferring, and collaboratively solving cases.

3. Users change what the technology means

The system becomes a shared knowledge and coordination infrastructure.

→ **Routines and technology develop together through repeated use, adaptation, and improvisation**

Orlikowski, W. J. (1996). Improvising organizational transformation over time: A situated change perspective. *Information systems research*, 7(1), 63-92.

A shared spreadsheet example

Context	Enacted as	Emerging practice	Consequence
Project team	Coordination tool	Everyone updates task status weekly	More transparency
Management	Control instrument	Managers monitor delays and ownership	More accountability pressure
Expert group	Flexible workaround	Users add hidden columns and acronyms	Local adaptation
Cross-functional team	Conflict arena	Teams dispute categories and definitions	Struggles over meaning

Exercise

Same technology, different use

Task:

- Choose one organizational technology, such as a project management tool, shared spreadsheet, customer database, or video-conferencing platform.
- Explain how two teams could use it differently, what routines could emerge, and what unintended change could follow.

Goal:

- Understand how technologies' effects emerge within contexts.

Modus:

- Pair work.

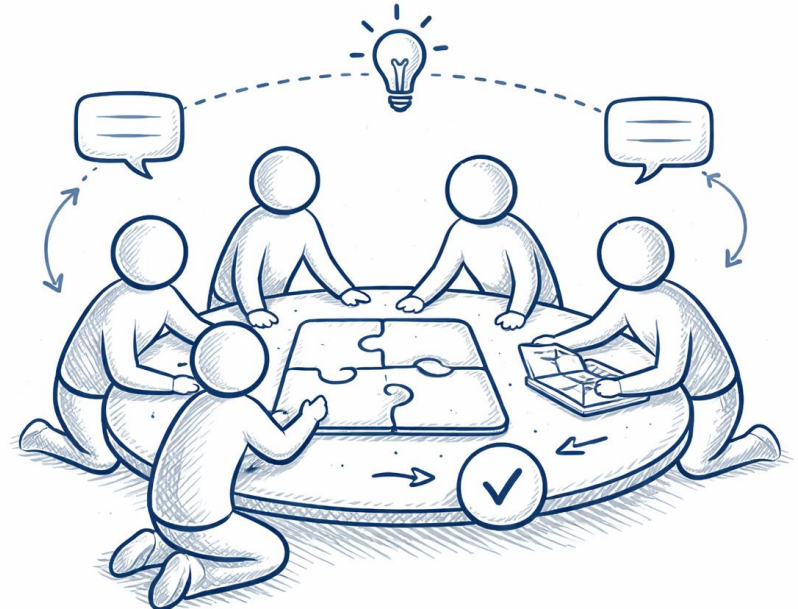


Image source: ChatGPT

Analytical toolkit to analyze technology in practice



Affordances

What possibilities for action does the technology offer in this context?

Technological frames

How do different groups interpret the technology?

Constraints

What does the technology make difficult, costly, or impossible?

Enactment

How is the technology used in practice?

Technology and control

Postmodern concerns with power and surveillance

- People may be judged by their contribution to system efficiency.
- Knowledge may count only when translated into information.
- Actors may struggle over control of information.
- Cyber-surveillance can expand organizational control.
- Technology can enable domination and resistance.

Exercise

Technology at TUM

Task:

- Analyze the core technologies TUM uses to guide students through an interpretative lens
- Questions
 - What are the core technologies that TUM uses?
 - How are these technologies enacted in practice?
 - How are these technologies interpreted differently across the organization?
- Summarize your insights on maximum two slides.

Goal:

- Understand how technologies matter in practice.

Modus:

- Group work.

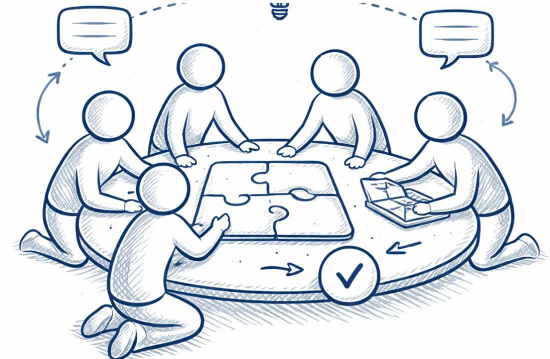


Image source: ChatGPT

Takeaways from today

Technology is not just a tool, but at the core of organizations.

Modernist theories classify technology by complexity, standardization, variability, and analyzability.

These classifications help explain organizational structure.

Symbolic perspectives show that technologies gain meaning through use.

The same technology can generate different routines in different contexts.

AI differs from conventional technology. Thus, we need to consider it in the context of organizations.

Course agenda

Date	Main theme
13.04.2026	Definitional basics of organization
20.04.2026	Formal organizational design
27.04.2026	Organization and environment
04.05.2026	Strategic organization design
11.05.2026	Technology and organization
18.05.2026	Information technology, AI, and organization
01.06.2026	Informal organization

Date	Main theme
08.06.2026	Informal organization
15.06.2026	Dynamic perspectives on organizing
22.06.2026	Guest lecture, Felix Ehrenfels, mmmake
06.07.2026	Trends in organization design
13.07.2026	Trends in organization design
17.07.2026	Review and wrap-up

Contact information



Dr. Christian A. Mahringer

TUM School of Management,
Campus Heilbronn

christian.mahringer@bwi.uni-stuttgart.de

Connect with me on [LinkedIn](#) for recent updates.

Generative AI was used to refine this presentation. All AI-generated content has been reviewed, and the author assumes full responsibility for its accuracy and integrity.